Project Plan: Scalable Multi-Tenant Database Service

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1. Goals

1. **On-demand tenant provisioning**: Create an isolated SQLite database per tenant via API in seconds, with no manual infrastructure work.
2. **Strong tenant isolation**: One database file per tenant (database-per-tenant model), so there is no cross-tenant data leakage and each tenant can be backed up, restored, or deleted independently.
3. **Horizontal scalability**: Scale to thousands of tenants by distributing databases across compute nodes, with stateless API and routing layers.
4. **Fully managed operations**: Automated backups, monitoring, health checks, and lifecycle management (create, suspend, delete).
5. **Cross-tenant schema migrations (P1)**: Apply versioned migrations safely across all or selected tenants, with tracking, rollback, and failure isolation.
6. **Offline synchronization (P2)**: Let clients keep a local SQLite copy, work offline, and sync changes back to the tenant DB with conflict resolution.

2. Features

- **Tenant management API**: POST /tenants, GET /tenants/{id}, DELETE /tenants/{id}, suspend/resume.
- **Query API**: Execute SQL (parameterized) against a tenant's DB via REST or a lightweight driver/SDK.
- **Tenant routing layer**: Maps tenant ID to the storage location or node holding the DB; caches the mapping.
- **Authentication and authorization**: Managed by Cognito
- **Resource limits**: Per-tenant quotas (DB size, request rate, query timeout).
- **Backup and restore**: Scheduled snapshots to object storage and point-in-time restore from snapshot plus WAL segments.
- **Observability**: Per-tenant metrics (latency, size, error rate), structured Cloudwatch logs and alarms.

P1: Schema Migrations

- **Migration registry**: Versioned SQL migration files stored centrally.
- **Version tracking**: A schema_migrations table inside each tenant DB, plus a central catalog of tenant versions.
- **Fan-out execution**: Run migrations across tenants in parallel batches with concurrency limits.
- **Safety mechanisms**: Pre-migration snapshot, dry-run mode, canary rollout (subset of tenants first), automatic halt on failure threshold.
- **Rollback**: Down-migrations or restore from the pre-migration snapshot.

- **Status dashboard/API**: Progress, failures, and retry per tenant.

P2: Offline Sync

- **Client sync SDK**: Local SQLite on device with a change-tracking mechanism (change log table or triggers).
- **Push/pull protocol**: Client pushes local changes since last sync token; server returns remote changes.
- **Conflict resolution**: Configurable strategy (last-write-wins by default, with per-table custom resolvers as an option).
- **Sync tokens and versioning**: Per-tenant monotonic change sequence for incremental sync.
- **Resumable and idempotent sync**: Safe retries on flaky connections.

AWS Resources

Layer	AWS Service	Purpose
API entry	**Amazon API Gateway** (or **ALB**)	Public REST endpoints, throttling, auth integration
Authentication	**Amazon Cognito** / Lambda authorizer	Tenant and admin identity, JWT validation
Control plane compute	**AWS Lambda**	Tenant provisioning, catalog operations, migration orchestration triggers
Data plane compute	**Amazon ECS on Fargate** (or EC2 Auto Scaling)	Runs the SQLite query service; each task hosts a set of tenant DBs
Database storage	**Amazon EFS** (shared, per-tenant files) or **EBS**with **S3** as source of truth	Durable storage of SQLite files; EFS enables any node to mount any tenant DB
Object storage	**Amazon S3**	Backups, snapshots, WAL/replication segments, migration scripts, sync payloads
Tenant catalog/metadata	**Amazon DynamoDB**	Tenant registry, routing map, schema versions, quotas, migration state
Orchestration	**AWS Step Functions**	Migration fan-out workflows with retries, canary, and halt-on-failure logic
Async processing	**Amazon SQS** (+ DLQ)	Migration job queue, backup jobs, sync processing
Scheduling	**Amazon EventBridge**	Scheduled backups, cleanup, health checks
Secrets/encryption	**AWS KMS**, **Secrets Manager**	Encryption at rest per tenant/key, credentials
Monitoring	**CloudWatch** (Logs, Metrics, Alarms), **X-Ray**	Observability and tracing